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Impact of cardiovascular and metabolic comorbidities on severity and outcomes of hospital-acquired sepsis in intensive care patients: a case–control study

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Abstract

Background Sepsis is a life-threatening dysregulated host response to microbial infection often leading to multiorgan failure and cardiovascular complications, which significantly contribute to mortality. This study aimed to identify key risk factors associated with disease severity among sepsis patients with distinct cardiovascular and metabolic comorbidities.

Methods This hospital-based case-control study included 148 adult patients with sepsis admitted to the intensive care unit (ICU) and 311 non-septic controls with comparable cardiovascular and metabolic comorbidities. Sepsis was diagnosed based on clinical criteria and confirmed by elevated serum procalcitonin levels. Serum procalcitonin, IL-6, IL-10, and TNF- α were measured using chemiluminescent immunoassays. Associations between comorbidities and sepsis severity were evaluated using chi-square tests and logistic regression and survival among sepsis cases was explored using Kaplan–Meier analysis.

Results Mostly infections were caused by Gram-negative bacteria, mainly *Klebsiella pneumoniae* and *Escherichia coli*. Elevated serum procalcitonin levels were used to support the diagnosis of sepsis. Compared with controls, sepsis patients showed significant increases in serum levels of IL-6, IL-10 and TNF- α , indicating an enhanced systemic inflammatory response. Statistical analysis showed that cardiovascular diseases, hypertension and type-2 diabetes were significantly associated with increased disease severity. Descriptive survival analyses among sepsis patients suggested lower survival probabilities in those with rheumatic heart disease and peripheral arterial disease.

Conclusions Cardiovascular and metabolic comorbidities particularly myocardial infarction and type 2 diabetes mellitus are strongly associated with sepsis requiring intensive care. Elevated inflammatory cytokine levels

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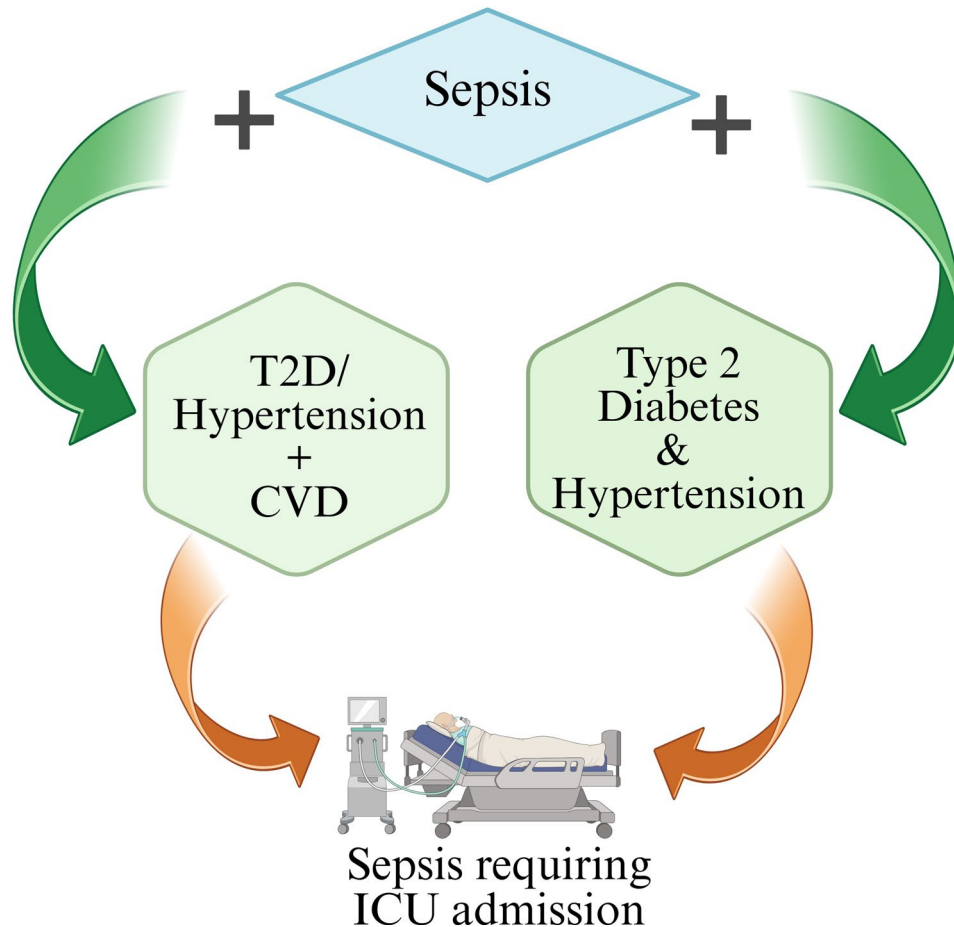


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characterize the systemic immune response in these patients. These findings highlight the importance of early risk stratification in septic patients with underlying cardiometabolic disease.

Keywords Sepsis, Cytokines, Cardiovascular diseases, Diabetes mellitus, Biomarker

Graphical Abstract



Introduction

Sepsis, a life-threatening organ dysfunction induced by an abnormal host response to microbial infection and one of the primary causes of mortality throughout the world [1]. Sepsis was responsible for 19.7% of all global mortality. The incidence and mortality decreased between 1990 and 2017 by 37% and 57%, respectively [2]. However, the burden of sepsis ranged considerably between regions and countries and highest incidence was observed in sub-Saharan Africa, Southeast Asia and Oceania [3].

The pathogenesis of sepsis has a complicated inflammatory component that involves the activation of both cellular (macrophage, endothelial, thrombocytic and cellular immunity) and plasmatic (complement activation, coagulation) systems. The innate immune response pushes the cytokine system to be activated for cytokine release [4]. This cytokine system has a wide range of effects on the vasculature and various organs, including

alterations in vascular permeability and endothelial function as well as the activation of additional mediators like bradykinin, histamine, the complement and the coagulation system. The pathophysiology of sepsis is significantly influenced by cytokines: anti-inflammatory cytokines (IL-10, TGF- β and IL-4) balance pro-inflammatory mediators such as TNF- α , IL-1, IL-6, IL-8, IL-12 and IFN- γ [5]. Particularly in secondary lymphoid tissue, homing chemokines aid in the organization of the adaptive immune system whereas inflammatory chemokines draw monocytes and PMNs to areas of inflammation and facilitate their passage through the blood vessel wall. Consequently, a number of inflammatory chemokines have been demonstrated to be more effective than IL-6 as possible biomarkers of sepsis. Among these are the chemokines monocyte chemoattractant protein (MCP-1) for the prediction of sepsis mortality and IL-8 for the diagnosis of sepsis [6]. Thus, inflammatory markers play

a crucial role in providing early signs of a grave prognosis in these patients.

Many studies suggest that sepsis survivorship is related to adverse long-term clinical outcomes, such as cognitive impairment, stress disorders, depression, increased risks of dementia and cardiovascular events [7]. Septic shock is one of the main subtypes of sepsis, which is characterized by chronic cardiovascular abnormalities and leads to a higher mortality rate of 40 to 50% [1]. Sepsis is a severe condition with major implications for cardiovascular health, particularly among patients with pre-existing cardiac conditions. Studies demonstrate that sepsis exacerbates underlying cardiac disease, leading to an increased risk of mortality and long-term cardiovascular complications [8]. Even though a smaller number of studies indicate the risk of cardiovascular patients during sepsis. The present case-control study aimed to evaluate the association between cardiovascular and metabolic comorbidities and sepsis severity, operationally defined as sepsis requiring ICU admission and to characterize the accompanying inflammatory profile using PCT, IL-6, IL-10, and TNF- α . PCT is widely used as a surrogate biomarker for bacterial infection and systemic inflammation, while IL-6 and TNF- α are key pro-inflammatory cytokines involved in sepsis-related organ dysfunction. IL-10 was included as a representative anti-inflammatory cytokine reflecting counter-regulatory immune responses. The objective of measuring these biomarkers in this study was descriptive and correlational, aiming to characterize the inflammatory profile in sepsis patients with cardiovascular and metabolic comorbidities, rather than establishing diagnostic or prognostic thresholds. By examining coexisting comorbidity patterns rather than isolated conditions and by focusing on a critically ill cohort from an under-represented population in eastern India. This study aims to provide clinically relevant insights into risk stratification and inflammatory burden in sepsis requiring ICU admission.

Materials and methods

Patients and control subjects

This hospital-based case-control study was with retrospective data collection conducted at a tertiary care hospital in Kolkata, India, between November 2021 and August 2022. A total of 459 participants were enrolled, including 148 cases and 311 controls. The comorbidities were documented prior to the diagnosis of sepsis based on medical records. Definitions of cardiovascular and metabolic comorbidities are provided in Supplementary Table 1. Sepsis was diagnosed according to the Sepsis-3 criteria, which defines it as life-threatening organ dysfunction caused by a dysregulated host response to infection. All cases had a clinically suspected or microbiologically confirmed infection and evidence of severe

organ dysfunction upon admission to the ICU as determined by treating intensivists based on standard clinical, laboratory, and hemodynamic parameters. Organ dysfunction was clinically assessed using indicators routinely available upon ICU admission. Disease severity was assessed using the Sequential Organ Failure Assessment (SOFA) score calculated at ICU admission.

Identification of causative organisms was performed using routine microbiological culture methods as part of standard clinical care. Blood, urine, and respiratory specimens were processed according to institutional protocols. Culture results were interpreted by clinical microbiologists in conjunction with clinical findings. Organisms considered contaminants were excluded based on routine clinical judgment. In cases of multiple organism isolation, the pathogen considered most clinically relevant was reported.

Controls were defined as individuals without sepsis, with similar cardiovascular and/or metabolic comorbidities, but without any clinical or laboratory evidence of active infection. Controls had normal serum procalcitonin levels and were not admitted to the ICU. They were recruited from non-ICU inpatient wards and outpatient departments during the same study period. Controls were not receiving antimicrobial therapy at the time of sample collection.

Written informed consent was obtained from all participants or their legally authorized representatives prior to enrolment. The study was approved by the Institutional Ethical Committee of BM Birla Hospital, Kolkata, and conducted in accordance with the Declaration of Helsinki.

Sample collection and processing

For sepsis cases, venous blood samples were collected within 24 h of ICU admission, following clinical diagnosis of sepsis and prior to or at the very early phase of definitive antimicrobial therapy. For control participants, blood samples were collected at the time of enrollment during routine clinical evaluation. Serum was separated immediately by centrifugation and stored at -80°C until analysis. All samples were thawed only once before testing. Commercially available kits were used according to the manufacturer's instructions, and the tests were performed in duplicate. Internal quality controls provided by the manufacturers were included in each test run. All comorbidities were recorded before the onset of sepsis, based on prior clinical diagnosis and treatment history documented in hospital records. However, Biomarkers (PCT, IL-6, IL-10, TNF- α) were measured after sepsis diagnosis for disease characterization and prognostic analysis. However, cytokines and PCT were biomarkers, not exposures. A schematic overview of patient selection,

inclusion and exclusion criteria, biomarker sampling and statistical analysis is shown in Fig. 1.

Inclusions and exclusions

Inclusion criteria were defined as adult patients (≥ 18 years) diagnosed with sepsis, admitted to the ICU and having at least one pre-existing cardiovascular and/or metabolic comorbidity. Sepsis was diagnosed based on clinical criteria and confirmed by elevated serum procalcitonin levels (>0.5 ng/mL). All cases included in the study had sepsis and required ICU admission. The cardiovascular comorbidities among cases included myocardial infarction (MI), coronary heart disease (CHD), atrial fibrillation (AF), heart failure (HF), rheumatic heart disease (RHD), coronary artery disease (CAD), and congestive cardiac failure (CAF). Additional comorbid conditions included diabetes mellitus, hypertension, cerebrovascular disease, deep vein thrombosis, diabetic foot, chronic kidney disease, chronic obstructive pulmonary disease, and rheumatoid arthritis. MI represents a history of acute coronary events, whereas CHD was used to denote chronic ischemic heart disease without documented infarction. The patients included in this study were those who developed infections 48–96 h after admission to the tertiary care hospital or within 5 days

after hospital discharge, consistent with healthcare-associated or hospital-acquired infection. A SOFA score ≥ 2 was used to identify patients with sepsis-associated organ dysfunction, reflecting clinically severe disease requiring ICU admission.

Patients were excluded if the infection was identified within the first 48 h of hospital admission, indicating community-acquired infection, or if the infection occurred more than 7 days after hospital discharge. Patients with incomplete clinical records, missing key demographic information, undocumented comorbidity status, or unavailable biomarker measurements (procalcitonin or cytokine assays) were excluded from the final analysis.

PCT assay

Sepsis diagnosis was established based on clinical assessment by the treating physicians in accordance with contemporary sepsis definitions, including evidence of infection and organ dysfunction requiring intensive care unit admission. Serum procalcitonin (PCT) was measured as a supportive biomarker of bacterial infection but was not used as a sole diagnostic criterion for sepsis. Patients with low or negative PCT values were included if they met clinical criteria for sepsis. Serum

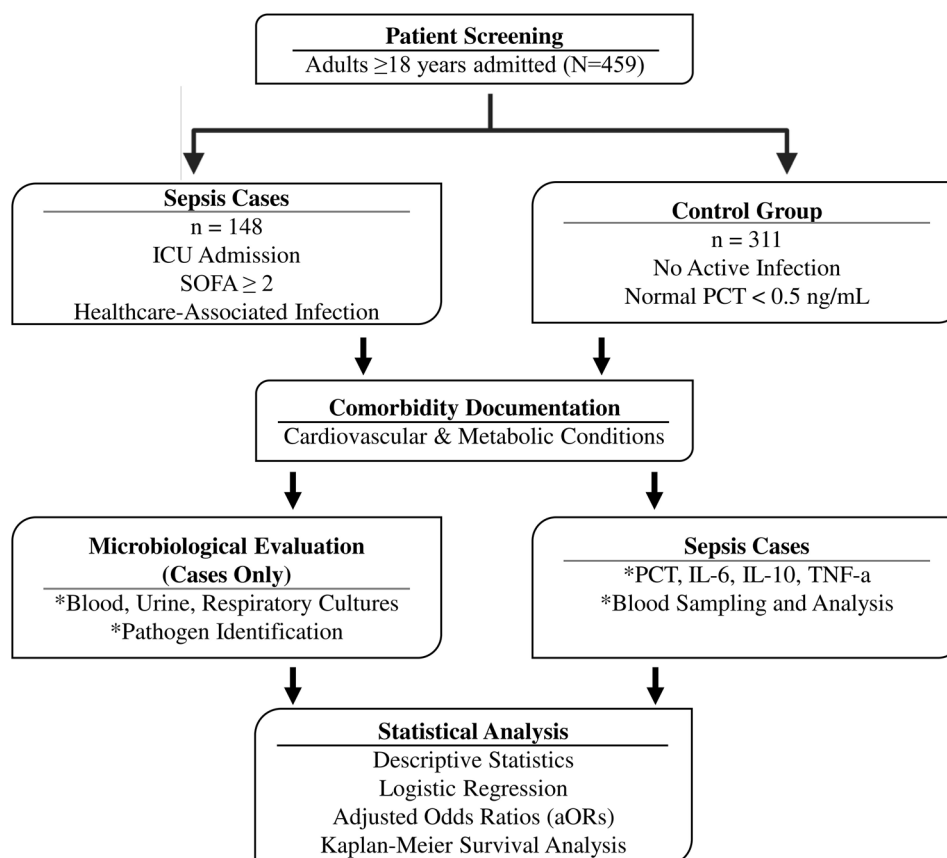


Fig. 1 Study workflow and participant selection

PCT concentrations were measured using the Elecsys BRAHMS PCT assay (non-competitive chemiluminescent immunoassay) on a cobas e411 analyzer. Samples from 148 sepsis cases and 311 controls were analyzed in parallel. The reference interval for PCT was <0.5 ng/mL.

IL6, IL10 and TNF α assays

The levels of IL6 were measured by the Roche IL-6 non-competitive chemiluminescent immunoassay. The test is said to be able to measure between 1.5 and 5000 pg/mL, have a limit of quantitation of 2.5 pg/mL, an inter-assay accuracy of 8.5% (at 17.3 pg/mL) and 2.9% (at 891 pg/mL). The samples were initially incubated with specific antibodies and then incubated again with specific antibodies labelled with ruthenium complexes. This process results in the formation of a sandwich complex. Subsequently, the complexes are trapped using magnetic forces and a voltage is applied to generate a chemiluminescent emission that is directly proportional to the concentration of IL6.

IL10 and TNF α were measured by the ELISA method. The qualitative immunoenzymatically determination of inflammatory markers in sepsis is based on the ELISA (Enzyme Linked Immunosorbent Assay) technique. Microtiter strip wells are pre-coated with corresponding antibodies of the specimen. After washing wells to remove all unbound sample material, horseradish Peroxidase (HRP) labeled anti-human IgM conjugate was added. This conjugate binds to the captured inflammatory marker-specific antibodies. The immune complex formed by the bound conjugate is visualized by adding Tetramethylbenzidine substrate that gives a blue reaction product. The intensity of this product is proportional to the quantity of inflammatory markers in the specimen. Sulphuric acid is added to stop the reaction. This produces a yellow end-point color. Absorbance of dual wavelength at 450/620 nm is read using an ELISA microwell plate reader.

Sample size calculation

The sample size was determined based on the number of eligible patients admitted to the intensive care unit

during the study period (November 2021 to August 2022). A formal prior sample size calculation was not possible due to the lack of reliable local epidemiological data on sepsis with specific cardiovascular and metabolic comorbidities. Therefore, a post-hoc power analysis was conducted using the observed proportions of the primary outcomes. The study had over 80% power to detect an odds ratio of 2.0 or greater for major comorbidities at a two-sided significance level of 0.05 with 148 sepsis patients and 311 control group participants. This sample size was considered sufficient to assess the associations between comorbidities and the severity of sepsis.

Statistical analysis

The statistical analyses were conducted using IBM Statistical Package for the Social Sciences (SPSS) v25. Since the dataset contains no outliers, duplicates, or missing values, data cleaning is not required. The methodology entailed computing the mean, standard deviation and percentage to estimate the proportion and average of the variables. The study performed chi-square testing (χ^2) to assess the occurrence of sepsis and patients with documented cardiovascular comorbidities in relation to the distribution of covariates. The odds ratios (ORs) and adjusted odds ratios (aORs) with 95% confidence interval (95% CI) were computed by univariate and multivariate logistic regression analysis in SPSS, respectively. The variables included in the multivariate logistic regression model were pre-selected based on biological plausibility, clinical significance, and to avoid multicollinearity and model overfitting. Age and sex were included as confounding factors, while type 2 diabetes, hypertension, myocardial infarction, and atrial fibrillation were included as primary outcomes. Finally, Kaplan-Meier Curve were used for survival analysis. Median value was used to divide the group into low and high level using median. All statistical analyses were conducted using a two-sided test and a p -value <0.05 was considered statistically significant.

Results

Patients characteristics

This comprehensive study comprised patients with sepsis who were admitted to the intensive care unit (ICU) and also had cardiovascular diseases. The study consists of 148 patients and 311 individuals as a control. In our dataset, the mean age of the patients was observed at 57 ± 16 years and 77 (52%) of patients are females and 71 (48%) are males ($p=0.622$) (Table 1). Predominantly, all the patients experience any one of the cardiovascular comorbidities, which are 34% (51) of the patients having myocardial infarction ($p<0.001$), 6.1% of patients having coronary heart disease ($p<0.001$), 15% (23) of cases occurring atrial fibrillation ($p<0.001$), 5.4% (8) patients having peripheral artery diseases ($p<0.001$) and 12.6%

Table 1 Characteristics of patients and control in the cohort

Variables	Category	Control (N=311)	Cases (N=148)	p-value
Age	<60 Years	135 (43.4%)	71 (48%)	0.358
	>60 Years	176 (56.6%)	77 (52%)	
Sex	Male	148 (47.7%)	71 (48%)	0.939
	Female	163 (52.4%)	77 (52%)	
Co-morbidity in cases	Yes	311 (100%)	148(100%)	<0.001
	No	0 (0%)	0 (0%)	
Survival in cases	Dead	0 (0%)	51 (34.5%)	<0.001
	Alive	311 (100%)	97 (65.5%)	

(19) cases having rheumatic heart disease ($p < 0.001$) in the patients who were admitted to the ICU. All sepsis cases received empirical antibacterial therapy as part of standard ICU management. In addition, 37% (55/148) of patients received antifungal therapy based on clinical suspicion or microbiological evidence of fungal infection. Among the cases, 34.5% (51) of patients were deceased and 65.5% (97) of cases were alive ($p < 0.001$) (Table 1). Microbiological culture identified pathogens in the majority of cases, with gram-negative organisms accounting for 84.5% (125/148) of infections. Predominantly, 49 patients were infected by *Klebsiella pneumoniae* and 42 patients by *Escherichia coli*. Additionally, *Acinetobacter baumannii* (13 patients) and *Pseudomonas aeruginosa* (8 patients) infections were found in the patients. Similarly, 16 patients were infected by gram-positive organisms, which included *Staphylococcus haemolyticus* (4 patients), *Candida tropicalis* (3 patients) and *Staphylococcus aureus* (2 patients) infections (Supplementary Table 2). The mean duration of ICU stay among sepsis patients was 8.04 ± 2.97 days.

All included patients met the Sepsis-3 criteria with sepsis-associated organ dysfunction. The mean SOFA score at ICU admission was 6.13 ± 3.11 .

Identification of sepsis

We performed a PCT assay and confirmed the sepsis using the Elecsys BRAHMS PCT non-competitive chemiluminescent immunoassay kit. We measured the procalcitonin level in 148 patients and 311 control individuals. We found the level of procalcitonin range from 0.8 to 57 ng/ml and the mean of 12 ± 11.65 ng/ml in the patient's serum and in the control range from 0 to 5 pg/ml and the average of 0.043 ± 0.28 ng/ml (Fig. 2a). From there, sepsis was confirmed based on the increased level of procalcitonin in the serum.

Evaluation of IL6, IL10 and TNF- α levels in serum

For estimation of IL6, IL10 and TNF- α levels in patients, we performed the Roche IL-6 non-competitive chemiluminescent immunoassay kit. The average concentration of IL6 in the patient's serum was observed at 38.15 pg/ml and the range was between 25 and 98 pg/ml. However, the IL6 level in control was found to be 12.84 pg/ml and in the range of 9–16 pg/ml. Similarly, the IL10 level ranged between 5 and 154 pg/ml and the mean of IL10 was found to be 39.73 pg/ml in the ICU patients. However, the mean was 3.54 pg/ml and the range were 1–5 pg/ml in control individuals. Finally, the mean concentration of TNF- α in ICU patients was observed at 26.8 pg/ml, with a range of 1–47 pg/ml. Meanwhile, the TNF- α level in control was 9.26 pg/mL (Fig. 2b).

Association between severity of patients with sepsis and co-occurring cardiovascular diseases and survival

In the unadjusted analyses (Table 2), several cardiovascular and metabolic conditions showed significant associations with sepsis. A lower rate of myocardial infarction, coronary heart disease, heart failure, and coronary artery disease was observed in sepsis patients compared to control groups, while invasive cardiac procedures such as coronary artery bypass grafting and coronary angiography were more common among sepsis patients. Conversely, metabolic disorders, particularly type 2 diabetes mellitus (T2D) and hypertension, were significantly more prevalent in sepsis patients. Among cardiovascular conditions, the prevalence of myocardial infarction (MI) was significantly lower among sepsis patients admitted to the ICU than among controls (OR = 0.516, 95% CI = 0.344–0.774). Similarly, coronary heart disease (CHD) was less common among sepsis cases (OR = 0.404, 95% CI = 0.191–0.852). In contrast, procedures such as coronary artery bypass grafting (CABG) and coronary angiography were significantly more common in the sepsis group (CABG: OR = 4.106, 95% CI = 2.056–8.200; angiography: OR = 6.162, 95% CI = 2.359–16.096). The

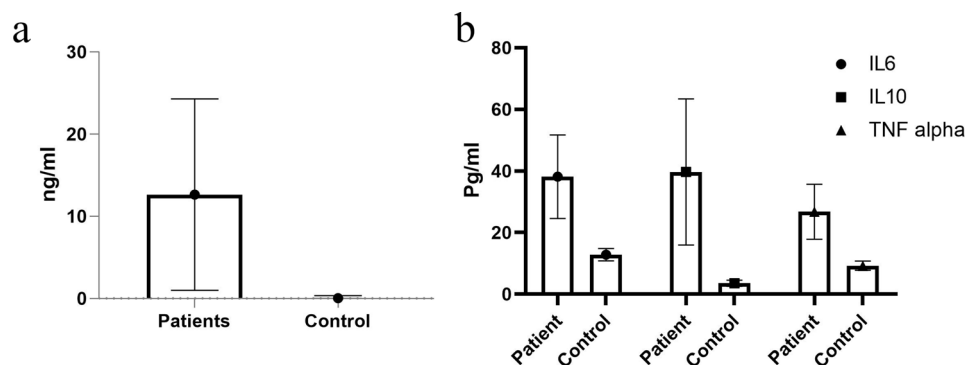


Fig. 2 (a) Level of Procalcitonin measured in sepsis and control groups. (b) Level of IL6, IL10, TNF alpha cytokines in sepsis patients who admitted in ICU and controls groups

Table 2 Univariable analysis of cardiovascular and metabolic comorbidities associated with severe sepsis (ICU admission) compared with controls

Variables	Category	Control (N= 311)	Cases (N= 148)	OR (95% CI)	p Value
Age	< 60 Years	135 (43.4%)	71 (48%)	0.832 (0.562–1.232)	0.358
	> 60 Years	176 (56.6%)	77 (52%)		
Sex	Male	148 (47.7%)	71 (48%)	0.985 (0.665–1.457)	0.939
	Female	163 (52.4%)	77 (52%)		
Myocardial Infraction	Yes	157 (50.5%)	51 (34.5%)	0.516 (0.344–0.774)	0.001
	No	154 (49.5%)	97 (65.5%)		
Coronary Heart Disease	Yes	43 (13.8%)	9 (6.1%)	0.404 (0.191–0.852)	0.014
	No	268 (86.2%)	139 (93.9%)		
Cerebrovascular Disease	Yes	12 (3.9%)	3 (2%)	0.516 (0.143–1.855)	0.302
	No	299 (96.1%)	145 (98%)		
Atrial fibrillation	Yes	74 (23.8%)	23 (15.5%)	0.589 (0.352–0.988)	0.43
	No	237 (76.2%)	125 (84.5%)		
Heart failure	Yes	41 (13.2%)	2 (1.4%)	0.090 (0.022–0.378)	< 0.001
	No	270 (86.8%)	146 (98.6%)		
Peripheral artery disease	Yes	14 (4.5%)	8 (5.4%)	1.212 (0.497–2.957)	0.672
	No	297 (95.5%)	140 (94.6%)		
Rheumatic heart Disease	Yes	31 (10%)	19 (12.6%)	1.330 (0.724–2.443)	0.256
	No	280 (90%)	129 (87.2%)		
Coronary artery bypass grafting	Yes	14 (4.5%)	24 (16.2%)	4.106 (2.056–8.200)	< 0.001
	No	297 (95.5%)	124 (83.8%)		
Coronary Artery disease	Yes	32 (10.3%)	6 (4.1%)	0.368 (0.151–0.902)	0.023
	No	279 (89.7%)	143 (95.9%)		
Coronary Angiography	Yes	6 (1.9%)	16 (10.8%)	6.162 (2.359–16.096)	< 0.001
	No	305 (98.1%)	132 (89.2%)		
Congestive cardiac failure	Yes	10 (3.2%)	2 (1.4%)	0.412 (0.089–1.906)	0.242
	No	301 (96.8%)	146 (98.6%)		
Subarachnoid haemorrhage	Yes	22 (7.1%)	1 (0.7%)	0.089 (0.012–0.670)	0.003
	No	289 (92.9%)	147 (99.3%)		
Myocardial infraction with T2D or Hypertension	Yes	19 (6.1%)	49 (33.1%)	7.607 (4.243–13.540)	< 0.001
	No	292 (93.9%)	99 (66.9%)		
Atrial fibrillation with T2D or Hypertension	Yes	13 (4.2%)	21 (14.2%)	3.790 (1.841–7.804)	< 0.001
	No	298 (93.9%)	127 (85.8%)		
T2D	Yes	42 (13.5%)	77 (52%)	6.946 (4.394–10.981)	< 0.001
	No	269 (86.5%)	71 (48%)		
Diabetic foot	Yes	8 (2.6%)	6 (4.1%)	1.600 (0.545–4.699)	0.388
	No	303 (97.4%)	143 (95.9%)		
Deep vein thrombosis	Yes	35 (11.3%)	4 (2.7%)	0.219 (0.076–0.628)	0.002
	No	276 (88.7%)	144 (97.3%)		
Hypertension	Yes	23 (7.4%)	44 (29.7%)	5.298 (3.051–9.200)	< 0.001
	No	288 (92.6%)	104 (70.3%)		
Chronic Obstructive Pulmonary Disease	Yes	26 (8.4%)	11 (7.4%)	0.880 (0.423–1.833)	0.733
	No	285 (91.6%)	137 (92.6%)		
Chronic Kidney Disease	Yes	18 (5.8%)	9 (6.1%)	1.054 (0.462–2.406)	0.901
	No	293 (94.2%)	139 (93.9%)		
Rheumatoid Arthritis	Yes	24 (7.7%)	2 (1.4%)	0.164 (0.038–0.703)	0.006
	No	287 (92.3%)	146 (98.6%)		

T2D – Type 2 diabetes, OD – Odds Ratio. Values are presented as number (percentage). ORs with 95% confidence intervals were calculated using univariate logistic regression. Age is expressed in years. Odds ratios represent the likelihood of being a case (sepsis requiring ICU admission) compared with a control (non-sepsis individual)

prevalence of heart failure was significantly lower among sepsis patients (OR=0.090, 95% CI=0.022–0.378), as was coronary artery disease (CAD) (OR=0.368, 95% CI=0.151–0.902). However, the prevalence of combined metabolic–cardiovascular comorbidities, including type 2 diabetes mellitus (T2D) and myocardial infarction with hypertension, was significantly higher in the ICU sepsis group compared with controls (OR=7.607, 95% CI=4.243–13.540). Similarly, atrial fibrillation with T2D or hypertension was more common among sepsis patients (OR=3.790, 95% CI=1.841–7.804). Among metabolic disorders, T2D showed the most significant association with sepsis requiring ICU admission (OR=6.946, 95% CI=4.394–10.981). Similarly, hypertension was significantly more common in the ICU sepsis group (OR=5.298, 95% CI=3.051–9.200). Comorbidities such as peripheral arterial disease, cerebrovascular disease, chronic obstructive pulmonary disease (COPD) and chronic kidney disease (CKD), did not differ significantly between groups ($p>0.05$ for all). In contrast, the sepsis group had a significantly lower prevalence of subarachnoid hemorrhage (OR=0.089, 95% CI=0.012–0.670) and rheumatoid arthritis was also less common (OR=0.164, 95% CI=0.038–0.703). Several cardiovascular conditions showed odds ratios below unity when comparing sepsis cases with controls. These findings reflect differences in comorbidity prevalence between the study groups and should not be interpreted as protective effects against sepsis.

Multivariate logistic regression analysis was performed to identify factors independently associated with sepsis (Table 3). In the adjusted model, type 2 diabetes mellitus was strongly associated with sepsis requiring intensive care unit admission (aOR=8.471, 95% CI: 5.118–14.022). Hypertension was also independently associated with a higher likelihood of developing sepsis (aOR=7.584, 95% CI: 4.062–14.159).

Survival analysis was conducted only among sepsis patients admitted to the intensive care unit and was intended solely for descriptive purposes. Kaplan–Meier curves were used to visualize in-hospital survival according to the presence of different cardiovascular

comorbidities. Overall, no statistically significant differences in survival were observed between the cardiovascular subgroups (Supplementary Fig. 1). Although patients with rheumatic heart disease and peripheral arterial disease showed lower survival probabilities, these differences were not statistically significant ($p=0.074$ and $p=0.086$, respectively; Fig. 3a and b). Kaplan–Meier curves were also generated to visualize survival among patients with myocardial infarction and atrial fibrillation (Fig. 3c and d). These analyses were exploratory and descriptive, and no conclusions regarding survival risk or prognosis were drawn. Serum IL-6 and TNF- α levels were analyzed to describe the inflammatory profile of sepsis. No survival or outcome-based regression analyses were performed using cytokine levels, and therefore no associations with mortality were inferred (Supplementary Fig. 2).

Discussion

Sepsis remains a major cause of morbidity and mortality worldwide, particularly among individuals with pre-existing cardiovascular and metabolic conditions [1]. In the present hospital-based case-control study, we characterized clinical features, inflammatory biomarkers and comorbidity profiles among sepsis patients admitted to the intensive care unit (ICU) and compared them with non-septic controls recruited during the same period. Rather than establishing causality our findings provide a descriptive overview of severe sepsis in a high-risk population. We observed markedly elevated levels of procalcitonin, IL-6, IL-10, and TNF- α among sepsis patients which reflect a heightened systemic inflammatory response consistent with sepsis pathophysiology. In addition, Gram-negative organisms particularly *Klebsiella pneumoniae* and *Escherichia coli* were the predominant causative pathogens in line with reports from tertiary care settings. While metabolic and cardiovascular comorbidities such as type 2 diabetes mellitus and hypertension were more frequently observed among patients with severe sepsis requiring ICU admission, these associations should be interpreted cautiously given baseline differences between cases and controls and the absence of matching. Collectively, these findings highlight the complex inflammatory and clinical landscape of severe sepsis in patients with chronic comorbidities, while underscoring the need for carefully designed prospective studies to clarify causal relationships.

The relationship between sepsis and pre-existing cardiovascular or metabolic diseases is increasingly recognized as bidirectional and synergistic. Patients with T2D and hypertension exhibit chronic endothelial dysfunction, low grade inflammation and impaired immune responses, which collectively increase susceptibility to severe sepsis [9–11]. Hyperglycemia stimulates

Table 3 Multivariable logistic regression analysis of factors associated with sepsis

Variables	aOR	95% Lower	95% Upper
Age	0.832	0.52	1.333
Sex	0.899	0.566	1.429
Diabetes mellitus	8.471	5.118	14.022
Hypertension	7.584	4.062	14.159
Atrial fibrillation	0.368	0.188	0.721
Myocardial Infarction	0.481	0.296	0.78

aOR – Adjusted Odds Ratio. aORs with 95% confidence intervals were calculated using multivariate logistic regression

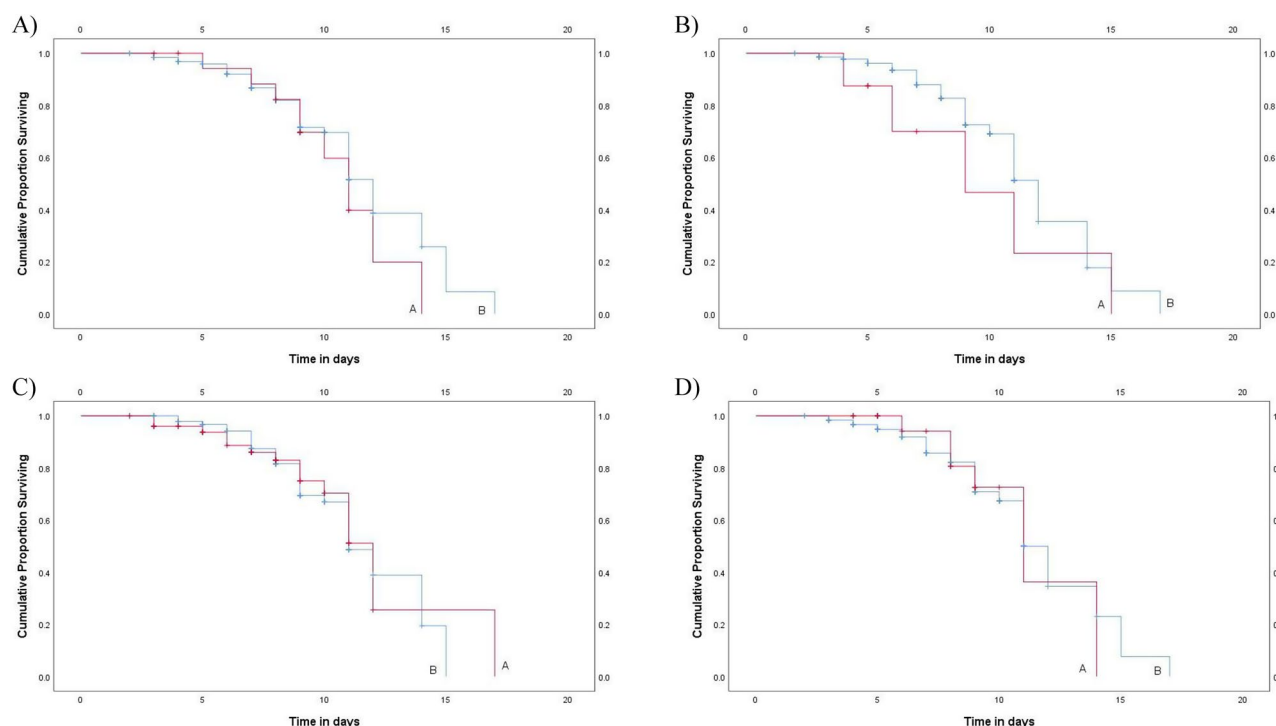


Fig. 3 Kaplan-Meier survival curves of sepsis patients admitted to the intensive care unit, stratified by the presence or absence of selected cardiovascular comorbidities, **(A)** Rheumatic heart disease (log-rank $p=0.074$); **(B)** Peripheral arterial disease (log-rank $p=0.086$); **(C)** Myocardial infarction (log-rank $p<0.001$); **(D)** Atrial fibrillation (log-rank $p=0.037$). The red line represents patients with the specified comorbidity, and the blue line represents patients without the comorbidity. Survival analysis was performed for descriptive purposes only

endothelial function, promotes leukocyte adhesion, cytokine production and microvascular injury all of which are key features of sepsis pathogenesis [12, 13]. Similarly, hypertension leads to vascular remodeling and microcirculatory dysregulation which exacerbates sepsis-induced hypotension and organ hypoperfusion [14]. In our study, the significant overrepresentation of T2D and hypertension among ICU sepsis patients supports previous epidemiological reports linking these comorbidities to poor sepsis outcomes [15–17].

The elevated IL-6, IL-10 and TNF- α levels observed in our cohort highlight the cytokine storm characteristic of severe sepsis. IL-6 is a pleiotropic cytokine that regulates acute phase responses, endothelial activation and leukocyte trafficking [18, 19]. Elevated IL-6 is strongly associated with sepsis severity and mortality [20, 21]. Similarly, TNF- α is a master regulator of inflammation which stimulates the release of other mediators, nitric oxide and reactive oxygen species, thereby causing tissue injury [22]. The association between increased IL-6/TNF- α levels and poor survival is consistent with studies showing these cytokines as independent predictors of septic shock and death [23, 6]. Elevation of IL-10 an anti-inflammatory cytokine represents a compensatory mechanism to counteract excessive inflammation [24]. However, prolonged IL-10 elevation may indicate immune exhaustion

or “sepsis-induced immunoparalysis” [25]. The interplay between pro- and anti-inflammatory mediators defines the path of sepsis; an imbalance can lead to persistent inflammation, secondary infections or multiorgan failure [26]. PCT is a biomarker produced by extrathyroidal tissues during bacterial infection [27, 28]. The significantly elevated PCT levels in our sepsis cohort reinforce its diagnostic value and reflect persistent systemic infection. These findings confirm previous reports demonstrating that elevated PCT bacterial burden is associated with sepsis severity and risk of mortality [29, 30].

In our cohort, Gram-negative bacteria accounted for 84.45% of infections, which is consistent with global trends in ICU sepsis [31, 32]. The predominance of Gram-negative sepsis may reflect higher endotoxin exposure and antibiotic resistance patterns [33]. *K. Pneumoniae* and *E. coli* are known to produce lipopolysaccharide (LPS) which induces Toll-like receptor 4 (TLR4) signaling and downstream cytokine production [34, 35]. LPS-induced systemic inflammation contributes to myocardial infarction, vascular leakage and multiorgan failure [36]. The high prevalence of these pathogens underscores the importance of early broad-spectrum empiric therapy followed by targeted T-squama therapy based on culture results.

In unadjusted analyses, the prevalence of myocardial infarction (MI), coronary heart disease (CHD), heart failure and coronary artery disease (CAD) was lower in sepsis cases than in controls resulting in odds ratios below unity. Importantly, these findings should not be interpreted as protective effects against sepsis. Rather, they likely reflect differences in baseline comorbidity burden and healthcare exposure between the study groups, particularly the higher prevalence of established cardiovascular disease among the control population. Similar paradoxical associations have been reported in observational studies where control groups are enriched for chronic disease or frequent healthcare utilization leading to counterintuitive odds ratios that reflect selection bias rather than biological protection [37–39]. Conversely, invasive cardiac procedures including coronary artery bypass grafting (CABG) and coronary angiography were significantly more frequent in sepsis cases. This observation reflects greater healthcare exposure, prior hospitalizations, and procedural interventions, which are recognized risk factors for hospital-acquired infections and subsequent sepsis [40, 41]. Patients with prior invasive cardiac procedures may have altered immune responses, endothelial dysfunction or prolonged exposure to intravascular devices, which increase susceptibility to severe infection [42].

In unadjusted and multivariate analyses, metabolic disorders emerged as strong independent predictors of sepsis requiring ICU admission. T2D showed the most pronounced association, followed by hypertension. These findings are consistent with large population-based and ICU cohort studies demonstrating that diabetes and hypertension significantly increase the risk of severe infection, organ failure and critical illness [43–46].

The biological plausibility of these associations is well-established. Chronic hyperglycemia in T2D impairs innate immune function through reduced neutrophil chemotaxis, phagocytosis, and oxidative burst activity, resulting in delayed pathogen clearance and prolonged inflammatory activation [47]. In addition, insulin resistance promotes endothelial dysfunction, oxidative stress and microvascular injury, which are central to the pathophysiology of sepsis-induced organ failure [48, 49]. Hypertension further exacerbates these mechanisms by inducing vascular stiffness, impaired autonomic regulation and increased endothelial permeability, thereby increasing tissue hypoperfusion and capillary leakage during sepsis [50].

Specifically, combined cardiometabolic conditions such as MI with T2D or hypertension and atrial fibrillation with metabolic syndrome were significantly more prevalent in sepsis cases. This observation underscores the overall burden of systemic dysfunction where metabolic, vascular and inflammatory abnormalities interact to

increase susceptibility to severe sepsis. Previous studies have similarly demonstrated that multimorbidity rather than isolated diagnoses is a major determinant of adverse outcomes in septic patients [51, 52].

Survival analyses were conducted exclusively among sepsis patients admitted to the ICU and were intended for descriptive purposes only. Kaplan-Meier curves did not show statistically significant survival differences across the cardiovascular subgroups. Although patients with rheumatic heart disease and peripheral artery disease had lower survival probabilities. However, the observed survival patterns are biologically plausible. Pre-existing cardiovascular disease may reduce physiological reserve, decreasing tolerance to the profound hemodynamic stress, myocardial depression, and hypoxia characteristic of severe sepsis [53]. Septic cardiomyopathy a reversible myocardial dysfunction occurring in up to 50–60% of septic patients may further compromise cardiac output, especially in patients with pre-existing ischemic or structural heart disease [54]. Arrhythmias such as atrial fibrillation can exacerbate this effect by reducing ventricular filling and promoting myocardial oxygen imbalance induced by tachycardia [55].

Limitations

The study was conducted at a single centre which may limit generalizability to a broader population. We lack longitudinal biomarkers of metabolic control and granular data such as HbA_{1c}, blood pressure history that could refine the interpretation of comorbidity severity relationships. ICU admission was therefore used as a surrogate marker of severe clinical presentation. Causality cannot be inferred from these cross-sectional data and residual confounding is possible. Microbiological identification was based on routine clinical diagnostics rather than standardized research protocols. Inflammatory biomarkers (PCT, IL-6, IL-10, and TNF- α) were measured simultaneously after sepsis diagnosis, which prevents the assessment of temporal trends or dynamic changes that could better reflect disease progression or prognosis. Accordingly, the biomarker findings should be interpreted as descriptive and correlative rather than predictive. Advanced molecular testing and systematic adjudication of polymicrobial infections were not performed, which may limit pathogen-level interpretation. Future multicentre studies, integrating dynamic biomarker profiling, cardiac imaging and immunophenotyping will help define the mechanistic pathways linking metabolic-cardiovascular dysfunction to sepsis severity. Furthermore, intervention trials evaluating whether early metabolic or cardiovascular optimization can alter sepsis outcomes in this high-risk group are warranted.

Conclusion

This study demonstrates that pre-existing cardiovascular and metabolic disorders are strongly associated with increased sepsis severity among critically ill patients admitted to the intensive care unit. Patients with conditions such as type 2 diabetes, hypertension and prior myocardial infarction exhibited significantly elevated inflammatory biomarkers including IL-6, IL-10, TNF- α , and procalcitonin compared to non-septic controls reflecting an amplified systemic inflammatory response. Gram-negative pathogens, particularly *Klebsiella pneumoniae* and *Escherichia coli* predominated in sepsis cases and may have contributed to the observed increased inflammatory burden. The co-occurrence of chronic cardiometabolic diseases may predispose patients to more severe manifestations of sepsis through mechanisms involving endothelial dysfunction, impaired immune responses and altered hemodynamic regulation. While survival analyses were exploratory, patients with certain cardiovascular disorders such as rheumatic heart disease and peripheral artery disease showed lower survival probabilities underscoring the potential clinical relevance of underlying vascular pathology in sepsis. These findings highlight the importance of early recognition of high-risk patients with cardiometabolic comorbidities, prompt infection management and vigilant monitoring in the critical care setting. Future multicenter, prospective and longitudinal studies incorporating standardized severity scores and dynamic biomarker assessment are needed to elucidate causal mechanisms and inform targeted strategies aimed at improving outcomes in septic patients with underlying cardiovascular and metabolic disease.

Abbreviations

ICU	Intensive care unit
PCT	Procalcitonin
ELISA	Enzyme-linked immunosorbent assay
CHD	Coronary heart disease
T2D	Type 2 diabetes
COPD	Chronic obstructive pulmonary disease
CKD	Chronic kidney disease
CVD	Cardiovascular disease
MI	Myocardial infarction
AF	Atrial fibrillation

Supplementary Information

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Supplementary Material 1

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Author contributions

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Data availability

All data supporting the findings of this study are available within the manuscript and its Supplementary Information.

Declarations

Ethics approval and consent to participate

The study was approved by the Institutional Ethical Committee of BM Birla Hospital, Kolkata, India. All research procedures were performed in accordance with the Declaration of Helsinki. Informed consent was obtained from all participants or their authorized representatives.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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